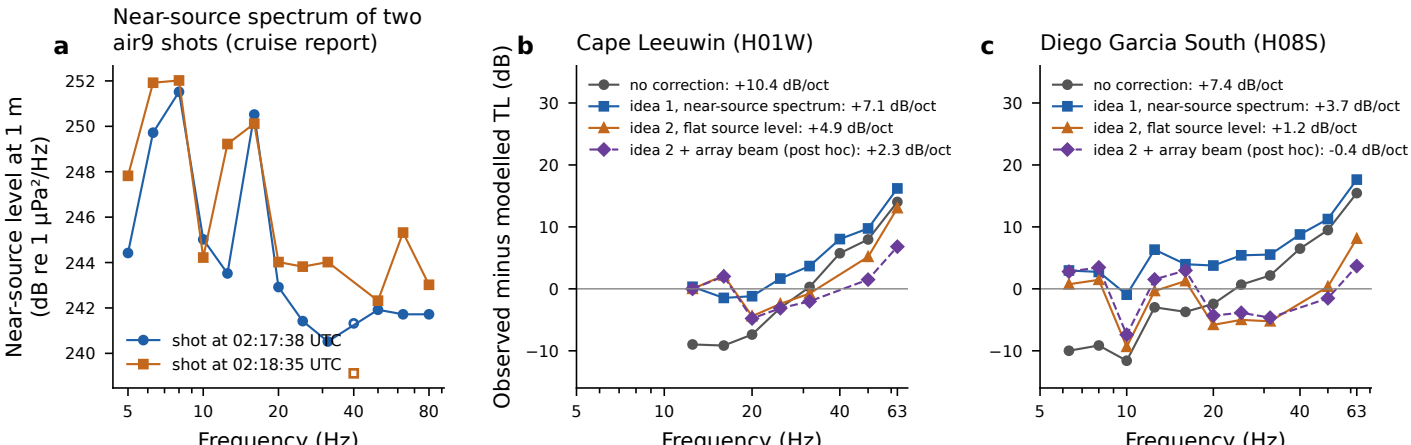
The pre-registered source corrections remove part of the airgun slope, not all of it



Plain-language note: The 2001 cruise report gives the depth of the airguns and a calibrated recording near the guns; a 1995 report gives the position and size of each of the 20 guns. We test two ideas (1 and 2) about the source level that the 2004 report used. If a correction were correct, its line in (b) and (c) would be flat. Neither tested idea is supported at Cape Leeuwin. A third combination (dashed) fits better, but we found it after we saw the data, so it needs a new test on other airgun sites.

Technical note: EW0112 pp. 5-7 (tow depth 11-12 m; Ball hydrophone calibrated, about 30 m ahead of the array, depth not stated); Fig. 2 DIGITISED (1/3-octave means; open markers < 5 % of full scale; 1-m scaling 24.7-26.8 m; 1-m peak 233 dB re 1 µPa). Array: USGS OFR 95-228 Fig. 4 (20 guns, 8470 cu in; EW0112 8465), strengths ∝ V^(1/3). r = Fig. 23 observed TL − module KRAKEN TL (M1, z_s 10 m). Idea 1: r = near-field factor at the Ball (x 30 m, z 15 m). Idea 2: r = C − S0. Pre-registered (ecbca309, d994b013): over 45 geometries both stations ≤ 3 dB/oct in 0 % (ideas 1 and 2): NOT SUPPORTED. Post hoc: + far-field array factor toward each station (12° off the tow axis; −0.2 dB at 12.5 Hz, −6.5 dB at 63 Hz): 78 %. Not a likelihood.